

Air Handling Unit Specifications

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Project: Lakefront Office Tower — Phase II

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SECTION 23 73 13 — CENTRAL-STATION AIR HANDLING UNITS

1.0 SCOPE OF WORK

Furnish and install factory-assembled central-station air handling units (AHUs) as indicated on mechanical drawings M2.01–M2.04 and as specified herein. Units shall serve the office floors (3–18), data center anteroom, and rooftop mechanical penthouse.

2.0 SUBMITTALS

- Shop drawings showing unit dimensions, weights, service clearances, and connection locations.
- Product data for each component: fans, coils, filters, dampers, and controls.
- AHRI-certified performance data including fan curves, coil capacity, and pressure drop.
- Operation and Maintenance (O&M;) manuals prior to substantial completion.
- Commissioning test reports per ASHRAE Guideline 1.5.

3.0 PERFORMANCE REQUIREMENTS

Tag	Service	CFM (Design)	ESP (in. w.g.)	Cooling (MBH)	Heating (MBH)	OA %
AHU-1	Low Floors 3–9	28,500	3.50	320	180	25%
AHU-2	High Floors 10–18	26,000	3.75	295	165	25%
AHU-3	Data Anteroom	8,400	2.25	120	60	15%
AHU-4	Penthouse MER	3,200	1.50	48	24	100%

4.0 EQUIPMENT SPECIFICATIONS

4.1 Cabinet Construction

Double-wall panel construction, minimum 2-inch insulation (R-13), 20-gauge G90 galvanized steel outer skin, 22-gauge painted inner liner. Thermal break construction required at all panel joints. Cabinet leakage $\leq 1\%$ of design airflow at 1-in. w.g. per ASHRAE 111.

4.2 Supply Fan Assembly

Parameter	Requirement
Fan Type	Airfoil backward-inclined centrifugal
Drive	Variable-frequency drive (VFD), NEMA Premium, IE3
Motor Enclosure	TEFC, 460V/3ph/60Hz
Bearing Life	L-10 = 200,000 hrs minimum (ABMA Std 9)
Vibration	≤ 0.10 in/sec RMS per AMCA 204
Isolation	2-inch deflection spring + neoprene pad

4.3 Cooling Coil

Chilled water cooling coil, 6-row minimum, copper tubes 5/8-in. OD on 1-1/2-in. triangular pitch, aluminum fins 12–14 FPI. Leaving air dry-bulb $\leq 55^{\circ}\text{F}$ at design conditions. Maximum face velocity 500 FPM. Provide drain pan sloped $\geq 1/8$ in./ft with insulated piping connection.

4.4 Heating Coil

Hot water heating coil, 2-row, copper tubes, aluminum fins. Entering water temp 140°F , leaving 120°F . Steam alternatives not permitted. Freeze-protection control sequence required per Section 23 09 23.

4.5 Filtration

Location	Filter Type	Efficiency	Initial ΔP	Final ΔP
Pre-filter	2-in. pleated panel	MERV 8	0.10 in.	0.50 in.
Final filter	4-in. rigid bag	MERV 13	0.25 in.	1.00 in.
Final (Data)	12-in. rigid box	MERV 15	0.35 in.	1.25 in.

5.0 SCHEMATIC DIAGRAM — AHU TYPICAL ASSEMBLY

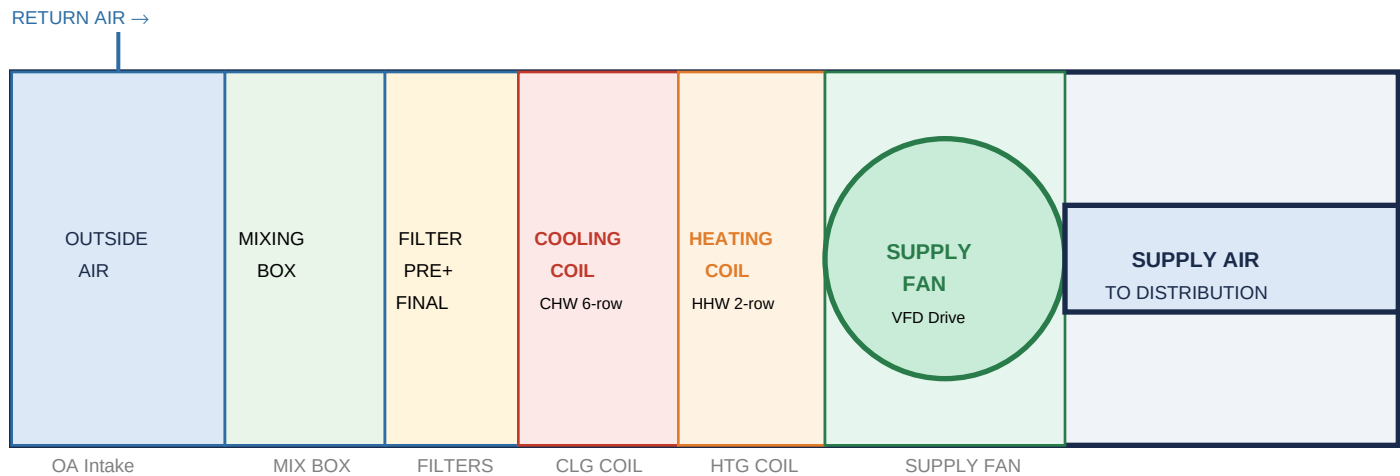


Figure 1 — Typical AHU Section Assembly (Not to Scale)

6.0 QUALITY ASSURANCE

Manufacturer shall have minimum 10 years of experience with commercial central-station AHUs. Units shall bear AHRI 430 and UL 1995 listings. Factory run-test each unit at full design conditions prior to shipment. Provide certified test reports with submittals.

7.0 WARRANTY

Manufacturer's standard 2-year parts and labor warranty. Extended 5-year warranty on fans and coil leakage. VFDs covered under separate drive manufacturer warranty, minimum 3 years.